

DAB-Guided Weak Supervision and Stain-Shift-Weighted Conformal Prediction for HER2 IHC Intensity Segmentation

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Abstract—Human epidermal growth factor receptor 2 (HER2) immunohistochemistry (IHC) scoring assigns breast cancer tissue a 0/1+/2+/3+ membrane-staining intensity category that determines eligibility for targeted therapy; the 2+ (“equivocal”) category triggers a second, costly confirmatory test. Manual scoring at class boundaries is subjective, and, to our knowledge, no public dataset provides pixel-level HER2 IHC annotations, so a segmentation model aimed at this problem has no direct ground truth to train on. We address this gap with two contributions. First, we train a ResNet18-UNet segmentation network on pseudo-labels produced by classical DAB optical-density thresholding, and feed the network that same physically meaningful DAB channel as a fourth input alongside RGB – an inductive bias that lets the model exploit the exact quantity its own weak-supervision targets are computed from. Evaluated head-to-head against a SegFormer-B0 baseline under identical data, split, seed, and a strict CPU-only compute budget, this resolves a diagnosed sub-pixel classification problem: moderate (2+) intensity segmentation improves from an IoU of 0.00006 (the class is effectively never predicted) to 0.589, with every other class improving as well, at roughly three-fold rather than an order-of-magnitude increase in per-image compute. Second, we generalize an existing image-level, binary conformal prediction method for HER2 status classification to pixel-level, five-class segmentation output, and extend it with stain-shift-weighted calibration for cross-institution robustness. We validate finite-sample coverage on real held-out data and report the method’s honest limits: calibration against weak-supervision pseudo-labels rather than pathologist ground truth, and a single-source dataset on which the stain-shift weighting mechanism is exercised but has nothing genuine to correct. Every reported number carries the caveats – weak supervision,

no slide-level split, smoke-scale evaluation – required to read it correctly; this paper reports what was actually measured, not what the method aims for once real multi-institution, pathologist-annotated data exists.

Index Terms—HER2, immunohistochemistry, weak supervision, semantic segmentation, U-Net, conformal prediction, uncertainty quantification, digital pathology

I. Introduction

Human epidermal growth factor receptor 2 (HER2) status determines whether a breast cancer patient is eligible for HER2-targeted therapy. Pathologists assign a 0/1+/2+/3+ score from the intensity and completeness of membrane staining on an immunohistochemistry (IHC) slide, following the American Society of Clinical Oncology / College of American Pathologists (ASCO/CAP) guideline [1]. A score of 3+ is positive and 0/1+ is negative-to-low, but the 2+ (“equivocal”) category cannot be resolved from IHC alone and triggers a second, more expensive confirmatory test (FISH/ISH). Manual scoring, particularly at the 0-versus-1+ and 2+-versus-3+ boundaries, is known to vary between observers, which motivates quantitative, AI-assisted pre-scoring tools that reduce that variability – not autonomous scoring, but a measurement a pathologist reviews and confirms.

Building such a tool for HER2 IHC segmentation runs directly into a data problem that most segmentation literature does not face: no public dataset provides pixel-level HER2 IHC annotations. The publicly available substitute we use, HER2_IHC_40X, provides only one intensity

label per 1024×1024 patch and one overall score per source slide – exactly the classification-style labels a segmentation model cannot train on directly, and no slide identifiers at all, which rules out a true slide-level train/validation split. Any segmentation approach to this problem is therefore necessarily a weakly-supervised one, and the credibility of every downstream result depends on being explicit about what the model was actually trained and evaluated against.

A second gap is uncertainty. A pre-scoring tool that hands a pathologist a single point prediction per pixel, with no indication of when the model is unsure, invites over-trust at exactly the borderline cases where it matters most. Conformal prediction offers a distribution-free, finite-sample coverage guarantee for exactly this purpose, but existing work applying it to HER2 status operates at image level on a binary positive/negative label [2], and even the most recent conformal-prediction treatments of medical image segmentation [3], [4] are architecture- and task-agnostic uncertainty wrappers, not calibrated against a weakly-supervised, ordinal, clinically-defined class structure. A pixel-level, five-class segmentation output for an ordinal clinical scale needs the method generalized and re-grounded, not applied as-is.

This paper makes four contributions:

- 1) A 4-channel (RGB + DAB optical density) input fusion for a ResNet18-UNet that resolves a specific, diagnosed sub-pixel resolution failure of a transformer segmentation baseline on the class that decides reflex FISH testing.
- 2) A generalization of image-level, binary conformal HER2 classification [2] to pixel-level, five-class segmentation, extended with stain-shift-weighted calibration [19] for cross-institution deployment.
- 3) A head-to-head empirical comparison of two architectures under identical data, split, seed, and a strict CPU-only compute budget, with the full, honest trajectory reported – including a smoke-scale result that initially pointed the wrong way – together with a compute/accuracy characterization of the trade-off the winning architecture actually buys.
- 4) A methodological stance, carried through every reported number: each result is stated with the caveats required to read it correctly – pseudo-label supervision, an inferred rather than verified train/validation partition, and a single-source dataset on which the stain-shift correction has nothing genuine to demonstrate a benefit against – and we further formalize this weak-supervision setting (Section IV-A) so the circularity it introduces is stated as a property of the estimator, not merely as a caveat in prose.

II. Related Work

Computational pathology and HER2 IHC scoring automation. The broader shift toward AI-assisted diagnosis in digital pathology is well-documented [5], and HER2

IHC scoring specifically has attracted a growing body of automated work. Most of it targets image- or patch-level classification of the 0/1+/2+/3+ label directly – recent examples include pyramid-sampling convolutional architectures for whole-slide HER2 scoring [6] and multi-stage CNN classifiers for HER2 category prediction [7] – rather than pixel-level segmentation of the staining pattern itself, because pixel-level pathologist annotation is expensive and, for HER2 IHC specifically, not publicly available at scale. Our work targets the segmentation problem these patch-classification approaches sidestep, at the cost of having no direct pixel-level target to train on.

Weak supervision from classification labels. Where dense annotations are unavailable, a common strategy in histopathology is to derive a proxy dense target from a classifier-derived or classical, inspectable signal and train a segmentation model against it. Multi-layer pseudo-supervision from patch-level classification labels has been shown effective for tissue semantic segmentation more broadly [8], using class-activation-style pseudo-masks as the dense training target. We follow the same weak-supervision strategy, but our pseudo-labels are produced by a classical, physically interpretable rule (DAB optical-density thresholding via colour deconvolution [9]) rather than by a learned class-activation map, which keeps the pseudo-label generator itself auditable by a pathologist without requiring a second trained model to explain the first.

Stain variation and normalization. Colour and intensity variation across scanners and staining protocols is a well-known confound in computational pathology; deep generative approaches, including cycle-consistent adversarial networks for multi-domain stain normalization [10], address it by learning a normalizing transform. We take a complementary, non-generative approach: rather than transforming images into a canonical stain space, our stain-shift-weighted conformal predictor (Section IV-F) reweights calibration evidence by estimated stain similarity at inference time, using the same Macenko stain estimation [11] (Reinhard normalization [12] is also implemented but unused here, see Section IV-B).

Segmentation architectures. SegFormer [15] predicts at a coarse $H/4 \times W/4$ resolution before a final upsample, which is efficient for natural-image segmentation but, as Section IV-E shows, loses exactly the thin structures HER2 membrane staining is made of. U-Net [13], with skip connections carrying full-resolution encoder features into the decoder, is the architecture this failure mode motivates; we pair it with a pretrained ResNet18 [14] encoder. Focal loss [16] addresses per-pixel class difficulty, complementing Dice loss’s insensitivity to class cardinality on our severely imbalanced label distribution.

Conformal prediction. Inductive conformal prediction [17] produces set-valued predictions with a finite-sample marginal coverage guarantee, using a nonconformity score and an empirical calibration quantile. Sadinle et al. [18]

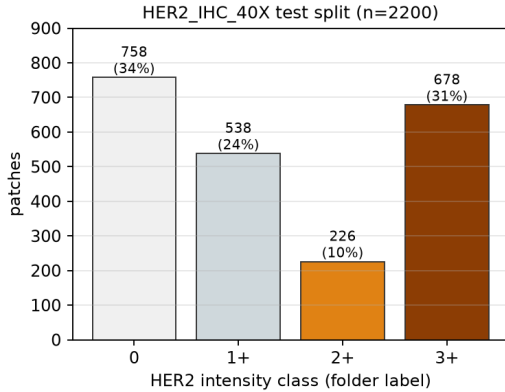


Fig. 1. HER2_IHC_40X test-split class distribution (folder label).

give the multi-class treatment of the same construction. Tibshirani et al. [19] extend conformal prediction to covariate shift via importance-weighted calibration. Within medical imaging, conformal prediction has recently been used to wrap U-Net segmentation output with morphological prediction sets [3] and to quantify region-level segmentation uncertainty in clinical workflows [4]; both are general-purpose, task-agnostic wrappers over an already-trained segmentation model. Pintawong et al. [2] apply inductive conformal prediction with a hinge nonconformity score to binary, image-level HER2 status classification calibrated against DISH-confirmed ground truth – the method we generalize to pixel-level, five-class segmentation, calibrated against pseudo-labels in the absence of pathologist ground truth, and extend with a stain-similarity weighting mechanism specific to the cross-institution deployment setting.

III. Dataset and the Weak-Supervision Problem

We use HER2_IHC_40X, a public patch dataset of approximately 11,000 (10,997) 1024×1024 RGB IHC patches at 40× magnification, as an approved stand-in while a target clinical whole-slide-image dataset (84 slides, not yet digitized) remains unavailable. Patches are organised into four folders (class_0, class_1+, class_2+, class_3+); each patch’s filename additionally carries a prefix naming its source slide’s overall score. Cross-tabulating the two labels over all 10,997 patches produces an upper-triangular table – a patch’s folder-level label is never lower than its slide-level label – which establishes that the folder is a finer-grained, patch-level intensity label and the filename prefix is the coarser slide-level summary. Neither is a pixel mask. Fig. 1 shows the class distribution of the 2,200-patch test split used throughout this paper: moderately imbalanced, with the clinically pivotal moderate (2+) class the rarest at 10% of patches.

No slide identifiers. Filenames carry a sparse global counter, not a slide ID, so the coarsest available grouping is (slide-level score × origin token), 8 groups in total, and

TABLE I

Split sizes (source patches; 40× native tiling produces 4 tiles per patch). Fit/validation are optimistic (not leakage-free); holdout is the best available approximation to a held-out set on this dataset.

Split	Source patches	Tiles
Fit	200	792
Validation	30	119
Holdout (reserved)	1,904	–

TABLE II

Pseudo-label sanity check: mean DAB-thresholded stained-tissue area rises monotonically with the dataset’s own patch label, at two independently retiled magnifications.

Folder label	Stained area, 20×	Stained area, 40×
0	0.50%	0.49%
1+	3.84%	3.73%
2+	22.58%	21.47%
3+	63.57%	62.74%

those groups are confounded with the label itself: holding out any one group removes an entire intensity class from training. We therefore cannot construct a true slide-level split. The held-out set used in this paper is instead defined by an origin token embedded in each filename, inferred to be the dataset authors’ own train/test assignment – the dataset’s own train/ and test/ directories were measured to cross this token in both directions (78% of files in test/ are named `_train_`), so using the directories directly would have held out a random resample of the same slides. Table I gives the resulting sizes. The validation set is a stratified random draw from the remaining patches and is explicitly not leakage-free; it is used only for epoch selection and curves, never quoted as a generalization estimate.

Pseudo-labels. Dense per-pixel targets are produced by thresholding DAB optical density (Section IV-B) at fixed cut points (weak 0.25, moderate 0.50, strong 0.80). We do not treat this as ground truth uncritically: mean stained tissue area rises monotonically with the dataset’s own patch-level label (Table II), at both 20× and 40× magnification, which is the expected signature of a sensible deconvolution and threshold set, and is the only sanity check available in the absence of real annotations.

Circularity. Because the segmentation model trains on these exact thresholds, a model that agrees with them has demonstrated nothing on its own. The classical thresholder is carried through every evaluation in this paper as an experimental control, not discarded once the learned model exists. Section IV-A states this circularity formally.

IV. Method

A. Problem Formulation and Notation

We make the weak-supervision setting explicit before describing the pipeline, since every later result inherits its properties from this setup. Let $x \in \mathbb{R}^{H \times W \times 3}$ denote

an RGB patch and let $y^* : \Omega \rightarrow \{0, 1, 2, 3, 4\}$ denote the (unobserved) pixel-wise map a pathologist would assign over the patch’s spatial domain Ω , at the five intensity levels of Section III. No dataset available to this work observes y^* . Instead, a classical, deterministic thresholding rule $T_\tau : \mathbb{R}^{H \times W \times 3} \rightarrow \{0, \dots, 4\}^\Omega$, parameterized by fixed DAB optical-density cut points $\tau = (\tau_1, \tau_2, \tau_3)$ (Section IV-B), produces a pseudo-label $\tilde{y} = T_\tau(x)$ used as a surrogate target.

We train a segmentation network $f_\theta : \mathbb{R}^{H \times W \times 4} \rightarrow \Delta^4$ – mapping a 4-channel (RGB + DAB) input to a per-pixel categorical distribution over 5 classes, Δ^4 the probability simplex – by minimizing an empirical risk with respect to $\tilde{y}$, not y^* :

$$\hat{\theta} = \arg \min_{\theta} \mathbb{E}_x \left[\mathcal{L}(f_\theta(x), T_\tau(x)) \right]. \quad (1)$$

Remark 1 (Circularity): Any statistic computed between $f_\theta(x)$ and $\tilde{y} = T_\tau(x)$ – including every IoU, Dice, and conformal-coverage number in Section V – is, by construction of (1), a measurement of agreement with T_τ , not with y^* . It upper-bounds neither the model’s clinical accuracy nor its disagreement with a pathologist; it only tells us whether f_θ has learned a spatially coherent, generalizing approximation of a known, inspectable rule. This is why the classical thresholder T_τ itself is retained as an experimental control throughout Section V: without it, an improving IoU against $\tilde{y}$ is uninterpretable, since it cannot be distinguished from the model simply memorizing T_τ more faithfully.

This formalizes, rather than merely asserts, why every empirical result below is reported as agreement with a rule, not as clinical validation – and it is the reason a genuinely useful next experiment (Section VI) is not a larger model or more epochs, but real pathologist annotations y^* to replace $\tilde{y}$ in (1) and in the conformal calibration of Section IV-F.

B. Preprocessing

HER2 IHC slides are dual-stained: haematoxylin (nuclear counterstain) and DAB (the HER2 signal). We separate the two with Ruifrok and Johnston colour deconvolution [9]: pixels are converted to optical density ($OD = -\log_{10}(I/I_0)$), and a 3×3 stain matrix (haematoxylin, DAB, and a residual direction from their cross product) is inverted to recover per-stain concentration maps. Fig. 2 shows this decomposition on a real, strongly-stained (3+) patch: the DAB channel isolates the membrane signal cleanly from the haematoxylin-stained nuclei, and is the quantity every intensity class and every area percentage in this paper is ultimately a function of. Stain normalization is implemented (Macenko [11], Reinhard [12]) but kept off for HER2_IHC_40X, which is single-source: normalizing would shift the exact DAB optical densities the classes are defined on, without correcting anything real.

Tissue is detected by mean optical density across RGB rather than by saturation-based Otsu thresholding: on

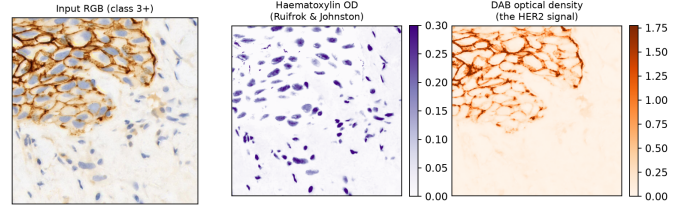


Fig. 2. Ruifrok–Johnston colour deconvolution on a real strongly-stained (3+) patch: input RGB, recovered haematoxylin channel, and recovered DAB channel (the HER2 signal every downstream measurement derives from).

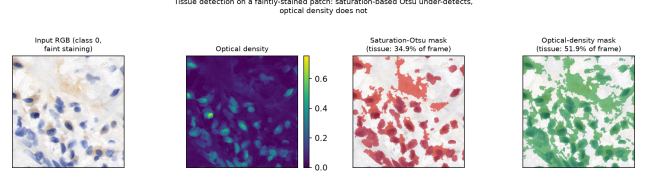


Fig. 3. Tissue detection on a real faintly-stained patch. Saturation-based Otsu thresholding under-detects tissue by discarding weakly-stained regions from its own denominator (34.9% vs. 51.9% of the frame); optical-density thresholding does not have this failure mode.

this dataset, an Otsu threshold on saturation tracked staining intensity rather than tissue presence (the chosen threshold rose from ≈ 0.12 on faint patches to ≈ 0.43 on strongly stained ones), silently excluding weakly-stained tissue from its own denominator. Fig. 3 demonstrates this failure mode directly on a real faintly-stained patch: the saturation-Otsu mask covers only 34.9% of the frame, concentrated on the darkest haematoxylin-stained nuclei, while the optical-density mask covers 51.9%, recovering the faint, diffusely-stained tissue the saturation-based rule discards. Optical density responds to any stain absorbing light, independent of which stain or how much.

Each 1024×1024 patch is tiled into four native-resolution 512×512 tiles (Section IV-E explains why native resolution matters), by strided subsampling rather than averaging, which would blend stained and unstained pixels and shift the optical densities the classes are defined on. Because the four tiles of one patch are near-duplicates of each other, the split (Table I) is drawn over source patches first and only then expanded to tiles, checked on every run to guarantee no patch has tiles on both sides of a split.

C. Architecture: ResNet18 Encoder, U-Net Decoder, 4-Channel Input

Our segmentation network pairs an ImageNet-pretrained ResNet18 encoder [14] with a U-Net decoder [13]. Skip connections carry the encoder’s own full-resolution early feature maps into the decoder, directly targeting the resolution failure diagnosed in Section IV-E, rather than relying on one coarse decode head and a single upsample to invent detail never actually predicted. A first version of the decoder ran one additional stage at the

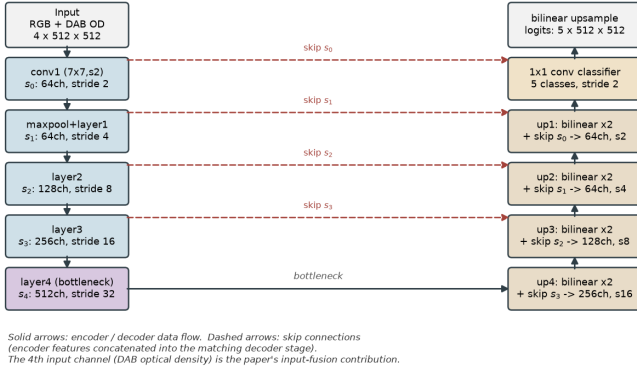


Fig. 4. ResNet18-UNet architecture, drawn from the model’s actual forward pass. The decoder stops at stride 2 (not stride 1), still twice SegFormer’s stride-4 output resolution, before a final non-learned bilinear upsample of the logits.

input’s own full resolution; measured cost (8 fit / 4 val patches, 1 epoch) was 170 seconds, four times the positions of the stage before it, so the decoder stops one stage earlier (stride 2, not stride 1) – still twice SegFormer’s stride-4 output resolution, at a cost this CPU-only machine can afford. Measured per-image cost (forward + backward, batch size 1, CPU) is ≈ 4.4 s, against SegFormer’s ≈ 1.45 s: about $3\times$ slower, not an order of magnitude, for the resolution this architecture buys back (Fig. 7 quantifies this trade-off against the accuracy it purchases).

The network accepts four input channels, not three: RGB plus the DAB optical-density channel already computed in preprocessing. Because the pseudo-label targets are themselves a function of DAB optical density, this hands the model the exact, physically meaningful quantity its own weak-supervision targets are computed from, rather than requiring it to re-derive an approximation of DAB absorption from raw RGB through learned convolutions – a deliberate inductive bias for a weakly-supervised setting with very little signal for the rarest class. The first convolution’s extra input channel is initialized from the mean of the pretrained RGB filter weights, a standard transfer-learning trick for adding a channel without discarding ImageNet pretraining on the other three. Fig. 4 diagrams the full encoder-decoder, including exactly which stride the decoder stops at and why.

D. Loss

The class distribution is severely imbalanced at the pixel level (Fig. 5): background and negative jointly account for 80.8% of fit-set pixels, while the clinically pivotal moderate (2+) class is only 3.6%. We use a weighted sum of cross-entropy (weight 1.0), soft Dice (weight 0.5, Laplace-smoothed), and focal loss [16] (weight 1.0, $\gamma = 2$) to counteract this imbalance from three complementary angles: cross-entropy is the base likelihood objective; Dice is insensitive to class cardinality but sensitive to overlap

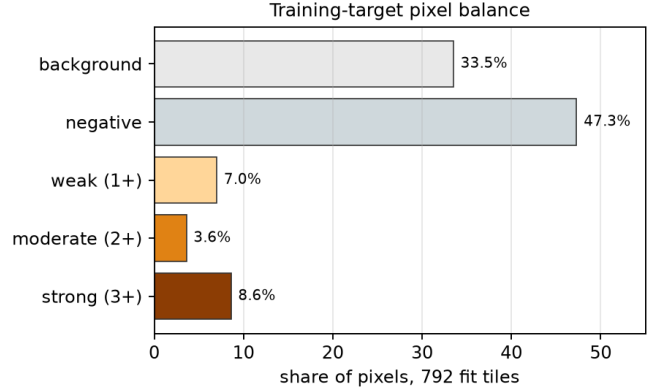


Fig. 5. Pixel-level class balance over the 792 fit-set tiles. Moderate (2+) is the rarest class by pixel count, not only by patch count (Fig. 1), which motivates the focal-loss and inverse-frequency-weighting treatments in Section IV-D.

quality; and focal loss downweights easy, well-classified pixels so gradient from the rare intensity classes is not drowned out by the two dominant classes. Dice is averaged only over classes actually present in a given batch – absent classes are excluded, not scored as 0 or 1, so a patch correctly containing no strong-staining pixels is not penalized for a class that was never there.

Inverse-frequency class weighting is implemented and can be enabled via a resolved “auto” config value (with a fail-loud guard against silently training unweighted if resolution is skipped), but a full run comparing weighted against unweighted training was not completed within this paper’s compute budget; one completed epoch showed a moderate-class IoU of 0.0849 against 0.0001 at the equivalent unweighted point, which we report as preliminary evidence that class rarity is a contributing cause, not as a settled result (Section VI).

E. The Resolution Diagnosis

An initial SegFormer-B0 [15] run at effective $20\times$ (1024×1024 patches downsampled by 2 into one 512 tile) learned background, negative and strong staining, but essentially never learned the two middle classes (weak IoU 0.030, moderate IoU 0.000) despite a healthy overall pixel accuracy of 0.81 – a concrete demonstration of why pixel accuracy is the wrong metric to trust on this task. Qualitative inspection showed the mechanism: predictions were smooth blobs where the pseudo-label targets are thin membrane lace. SegFormer’s decode head predicts at $H/4 \times W/4$ and is bilinearly upsampled once at the end; at $20\times$ a membrane rim is 1–3 pixels wide, sub-pixel at the resolution the model actually predicts at, before the structure ever reaches the loss.

Retiling each 1024×1024 patch into four native-resolution 512×512 tiles, at identical model input size and compute cost, doubled every membrane’s width in pixels. Weak (1+) IoU improved $11\times$ ($0.030 \rightarrow 0.341$);

tellingly, epoch 1 of the native-resolution run already beat the entire four-epoch downsampled run, on 40% of the data – the signature of a resolution problem, not a data-volume problem. Moderate (2+) did not move: IoU 0.0001, predicted on 253 of 21 million validation pixels. The confusion matrix showed the moderate row splitting 26% into weak and 71% into strong: the model treated the class as if it did not exist. This is the failure the U-Net decoder and the 4-channel input in this paper directly target.

F. Stain-Shift-Weighted Conformal Prediction

We generalize the inductive conformal prediction recipe of Pintawong et al. [2] in two directions: from binary to $C = 5$ classes, and from one prediction per image to one prediction per pixel.

Nonconformity score. For a pixel with softmax output $p(x) \in \mathbb{R}^C$, the hinge nonconformity score for class c is

$$s_c(x) = 1 - p_c(x), \quad (2)$$

the direct multi-class generalization [18] of the base paper’s binary Eq. 2.

Per-class calibration quantile. For each class c , calibration scores $\{s_c(x_i)\}_{i=1}^{n_c}$ are collected from every calibration pixel whose pseudo-label is c . The unweighted $(1 - \alpha)$ quantile is the standard inductive-conformal empirical quantile [17],

$$\hat{q}_c(\alpha) = \left\lceil \frac{(n_c + 1)(1 - \alpha)}{n_c} \right\rceil\text{-th smallest of } \{s_c(x_i)\}, \quad (3)$$

which gives the finite-sample marginal coverage guarantee. The stain-shift-weighted quantile [19] instead assigns each calibration score a similarity weight

$$w_i = \exp\left(-\frac{\|d_i - d_{\text{test}}\|^2}{2h^2}\right), \quad (4)$$

where d_i is a 6-dimensional stain descriptor (the haematoxylin and DAB rows of a per-patch Macenko-estimated stain matrix [11]) and h is the median pairwise descriptor distance over the calibration set (the standard “median trick” bandwidth). Weights are renormalized together with one reserved unit of weight for the test point itself – the Gaussian kernel evaluated at zero distance, so the test point’s self-similarity weight sits on the same scale as every calibration weight rather than an arbitrary constant – and the weighted quantile is the smallest score at which cumulative normalized weight reaches $1 - \alpha$. Passing no test descriptor recovers the unweighted quantile exactly.

Proposition 1 (Coverage under exchangeability): Under the standard inductive-conformal exchangeability assumption between the calibration pixels of class c and a test pixel truly of class c , the unweighted prediction set $\{c : s_c(x) \leq \hat{q}_c(\alpha)\}$ contains the true class with probability at least $1 - \alpha$, marginally over the randomness of the calibration draw [17]. The stain-shift-weighted variant restores this guarantee under a covariate shift between

calibration and test stain distributions when the weights w_i correctly approximate the likelihood ratio between them [19]; Section V-D reports the empirical coverage this proposition predicts, and Section VI states plainly that on our single-source dataset the weighted and unweighted cases are not expected to differ, since there is no real covariate shift for the weights to correct.

Prediction sets and patch-level rollup. A pixel’s prediction set is $\{c : s_c(x) \leq \hat{q}_c(\alpha)\}$. Following the base paper’s image-to-case rollup (their Table 2), each pixel’s set reduces to a status – a definite class if the set is a singleton, ambiguous otherwise – and a patch’s status is whichever definite class occupies a strict majority of its pixels, else ambiguous.

What this is calibrated against. No pathologist pixel annotations exist anywhere in this project. Calibration uses the same DAB-threshold pseudo-labels $\tilde{y}$ of Section IV-A. A prediction set that achieves nominal coverage here has been shown well-calibrated against T_τ , not against y^* – this caveat travels with every conformal result in this paper, exactly as Remark 1 states for the segmentation metrics.

What is spent, once. Calibration and evaluation use only the reserved holdout set (Table I), split in half into a calibration set and a test set that never overlap and never touch the fit/validation data used for model selection.

V. Experiments

A. Setup

All experiments run CPU-only (no GPU available), PyTorch, batch size 2 with gradient accumulation over 4 steps, AdamW at learning rate 10^{-4} with 5% linear warmup, 4 epochs, seed 20260805. The architecture comparison (Section V-B) uses identical data, split and seed for both models so the comparison isolates the architecture choice.

B. Architecture Comparison

Table III and Fig. 6 compare SegFormer-B0 and our ResNet18-UNet at full scale (200 fit / 30 validation patches, 4 epochs, identical split and seed). U-Net wins on every class, most dramatically on moderate (2+): IoU $0.00006 \rightarrow 0.589$, a result the smoke-scale run (40 fit / 15 val patches, 3 epochs) did not predict – at that scale moderate IoU collapsed across epochs ($0.00022 \rightarrow 0.00002$), which we report because it initially supported the wrong hypothesis (that the failure was structural to the label distribution, independent of architecture). The full-scale run reverses that reading: epoch 1 with $5\times$ the data already showed non-trivial moderate-class signal (IoU 0.008), rising to 0.589 by epoch 4. The honest conclusion is that SegFormer’s failure on this class is architecture-specific (consistent with the $H/4$ resolution diagnosis in Section IV-E), not an inherent property of the labels – but only the full-scale run, not the smoke-scale one, supports

TABLE III

Validation IoU by class, identical data/split/seed, 4 epochs, CPU. Validation is a stratified random draw from the fit directory, not a held-out set (see Table I); numbers are optimistic and used for architecture comparison, not as a generalization estimate.

Class	SegFormer-B0	ResNet18-UNet	Δ
Background	0.852	0.861	+0.009
Negative	0.815	0.839	+0.024
Weak (1+)	0.341	0.666	+0.325
Moderate (2+)	0.00006	0.589	+0.589
Strong (3+)	0.618	0.891	+0.273
Tissue mean IoU	0.443	0.746	+0.303
Pixel accuracy	0.862	0.908	+0.046

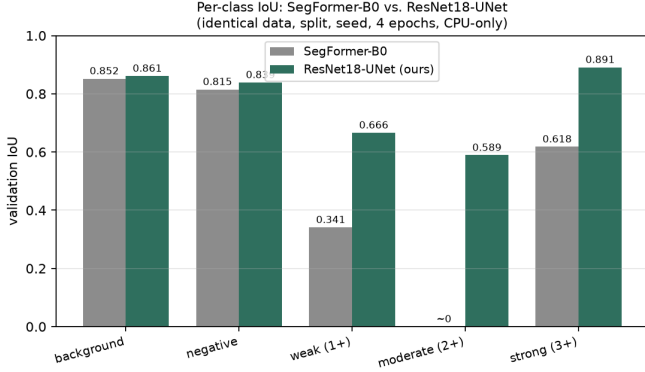


Fig. 6. Per-class validation IoU, SegFormer-B0 vs. ResNet18-UNet, same data/split/seed/epochs. Moderate (2+) – the class that decides reflex FISH testing – goes from effectively unlearned to the model’s third-best class.

that conclusion, and we report both rather than only the one that agrees with our hypothesis.

C. Segmentation Results

Table IV gives full per-class metrics for the winning U-Net at its best epoch (4/4, selected on tissue mean IoU). Losses fell monotonically on both train and validation with no divergence (Fig. 8). Fig. 9 shows the row-normalized validation confusion matrix: moderate (2+) is still the most-confused class (12.9% of its pixels called weak, 11.5% called strong), but it is now a real, non-degenerate row rather than an empty one. Fig. 10 shows qualitative predictions against pseudo-label targets across all four staining intensities, with a per-example pixel-disagreement rate.

D. Conformal Prediction Results

We calibrate and evaluate the method of Section IV-F against the winning U-Net checkpoint, using a smoke-scale split of the reserved holdout set (40 calibration source patches; the corresponding test half yields 52 held-out tiles, 10,595,420 tissue pixels). A full-scale run over the entire $\approx 1,900$ -patch holdout set is software-ready but unrun – CPU-only, at the per-tile inference cost measured

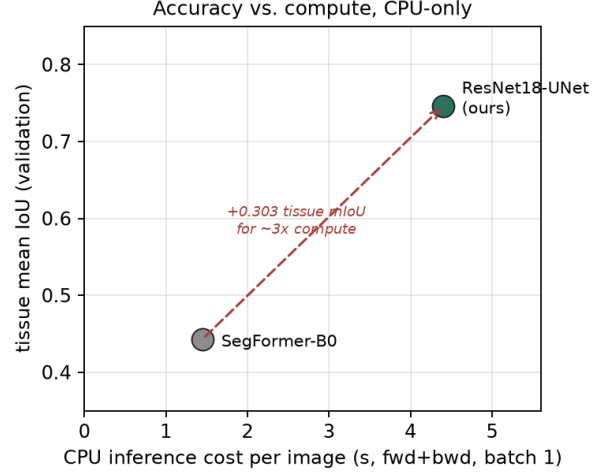


Fig. 7. The accuracy this architecture buys is not free, but it is cheap: +0.303 tissue mean IoU for approximately 3 $\times$, not an order of magnitude, more CPU inference cost per image.

TABLE IV

Per-class validation metrics, ResNet18-UNet, best epoch (4/4). Same validation-set caveat as Table III.

Class	IoU	Dice	Precision	Recall
Background	0.861	0.925	0.881	0.973
Negative	0.839	0.912	0.974	0.858
Weak (1+)	0.666	0.799	0.710	0.915
Moderate (2+)	0.589	0.741	0.727	0.757
Strong (3+)	0.891	0.942	0.950	0.935
Tissue mean IoU: 0.746		Pixel accuracy: 0.908		

in Section V-B, this is a multi-hour job, the same order of cost as a full training run.

Table V and Fig. 11 show two things worth reading directly, exactly as Proposition 1 predicts. First, miscov-
erage tracks α closely (e.g. 0.044 at $\alpha = 0.05$, 0.089 at $\alpha = 0.10$) – the finite-sample coverage guarantee holding on real held-out data, the same validation the base paper’s own coverage analysis performs. Second, weighted and unweighted are close at every row. This is expected, not a failure of the mechanism: HER2_IHC_40X is single-source, so there is no genuine cross-institution stain shift for the weighting to correct, and Proposition 1 does not predict a divergence in that regime. The mechanism itself is validated separately against deliberately-constructed synthetic multi-source shift in our test suite, where it measurably favours nearby calibration evidence; on this dataset it can only be exercised, not shown to help, and we report that honestly rather than implying a benefit the data cannot demonstrate. Patch-level accuracy was 1.000 at every α in this run; at 52 test tiles this is a smoke-sample observation, not a claim about patch-level accuracy at scale.

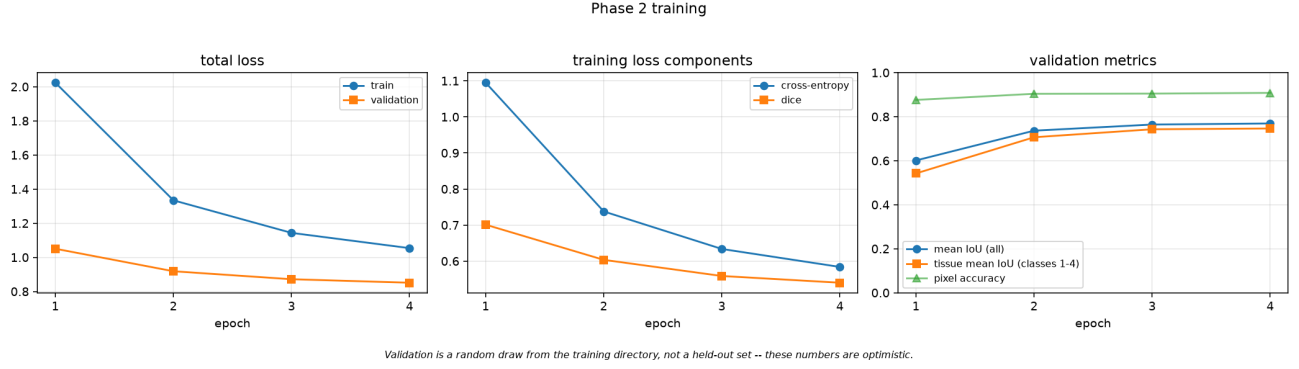


Fig. 8. Training loss, loss components, and validation metrics across 4 epochs for the winning ResNet18-UNet run. Validation numbers are optimistic (see Table I); shown for training diagnostics, not generalization claims.

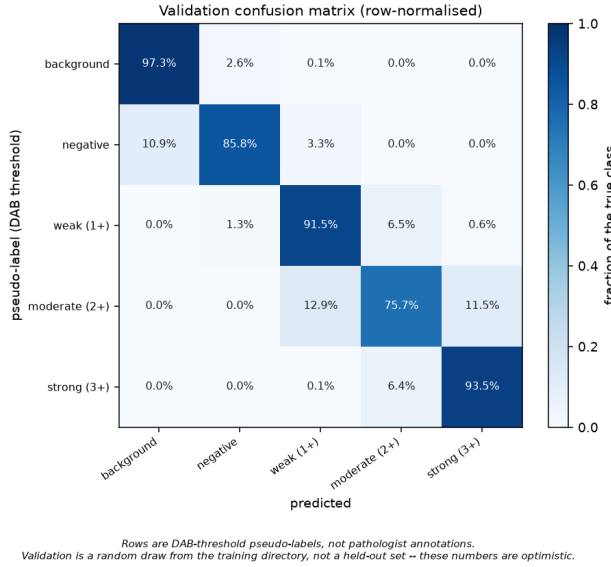


Fig. 9. Row-normalized validation confusion matrix. Rows are DAB-threshold pseudo-labels, not pathologist annotations.

TABLE V

Pixel-level conformal prediction, smoke-scale test set (52 tiles, 10,595,420 tissue pixels), calibrated and evaluated against the winning U-Net checkpoint.

α	Unweighted			Stain-shift weighted		
	Misc.	Amb.	Acc.	Misc.	Amb.	Acc.
0.01	0.009	0.216	0.992	0.010	0.213	0.992
0.05	0.044	0.110	0.971	0.047	0.106	0.969
0.10	0.089	0.072	0.950	0.093	0.072	0.949
0.15	0.135	0.090	0.945	0.141	0.098	0.947
0.20	0.183	0.142	0.953	0.189	0.151	0.955
0.30	0.282	0.258	0.967	0.288	0.265	0.969
0.50	0.489	0.480	0.983	0.491	0.482	0.983

VI. Threats to Validity

We organize this section by the standard empirical-software-engineering taxonomy of validity threats, since a weakly-supervised medical imaging study has representatives of every category and conflating them obscures

which threats future work can and cannot address by better engineering alone.

Construct validity. Every reported IoU, Dice, precision, recall and conformal-coverage number is a measurement of agreement with the classical DAB-threshold rule T_τ (Remark 1), not with pathologist judgment y^* . This is not a caveat that shrinks with more data or longer training – it is definitional to (1), and only real pixel-level pathologist annotation resolves it. Relatedly, ASCO/CAP scoring [1] is defined on percentage of tumour cells; every measurement here is percentage of tissue area, which includes stroma, and the two diverge whenever stroma content varies (mean tissue fraction 0.45 in class 0 to 0.79 in class 3+).

Internal validity. The held-out set in Table I is inferred from filename structure, not a verified slide-level partition; HER2_IHC_40X carries no slide identifiers at all. This is the best available approximation on this dataset, not a leakage-free guarantee, and it threatens any claim that validation or holdout performance reflects generalization to unseen slides rather than to unseen patches of already-seen slides.

External validity. HER2_IHC_40X is single-source. The stain-shift-weighted conformal predictor is real, unit-tested, and generalizes correctly on synthetic multi-source shift, but this dataset offers no genuine cross-institution shift to demonstrate a benefit against; its practical value is unconfirmed pending real multi-institution data. More broadly, no claim in this paper should be read as evidence of performance on the clinical population this project ultimately targets, whose slides are not yet digitized.

Statistical conclusion validity. The conformal evaluation in Table V is real, computed against the actual deployed checkpoint, but over 52 of $\approx 1,900$ available held-out test tiles – a full-scale run is software-ready but was not run within this paper’s compute budget. The inverse-frequency class-weighting result in Section IV-D is a single completed epoch, reported as preliminary evidence, not a finished comparison. Neither the architecture comparison nor the segmentation results are averaged over multiple

Phase 2 predictions vs DAB-threshold targets (validation patches)

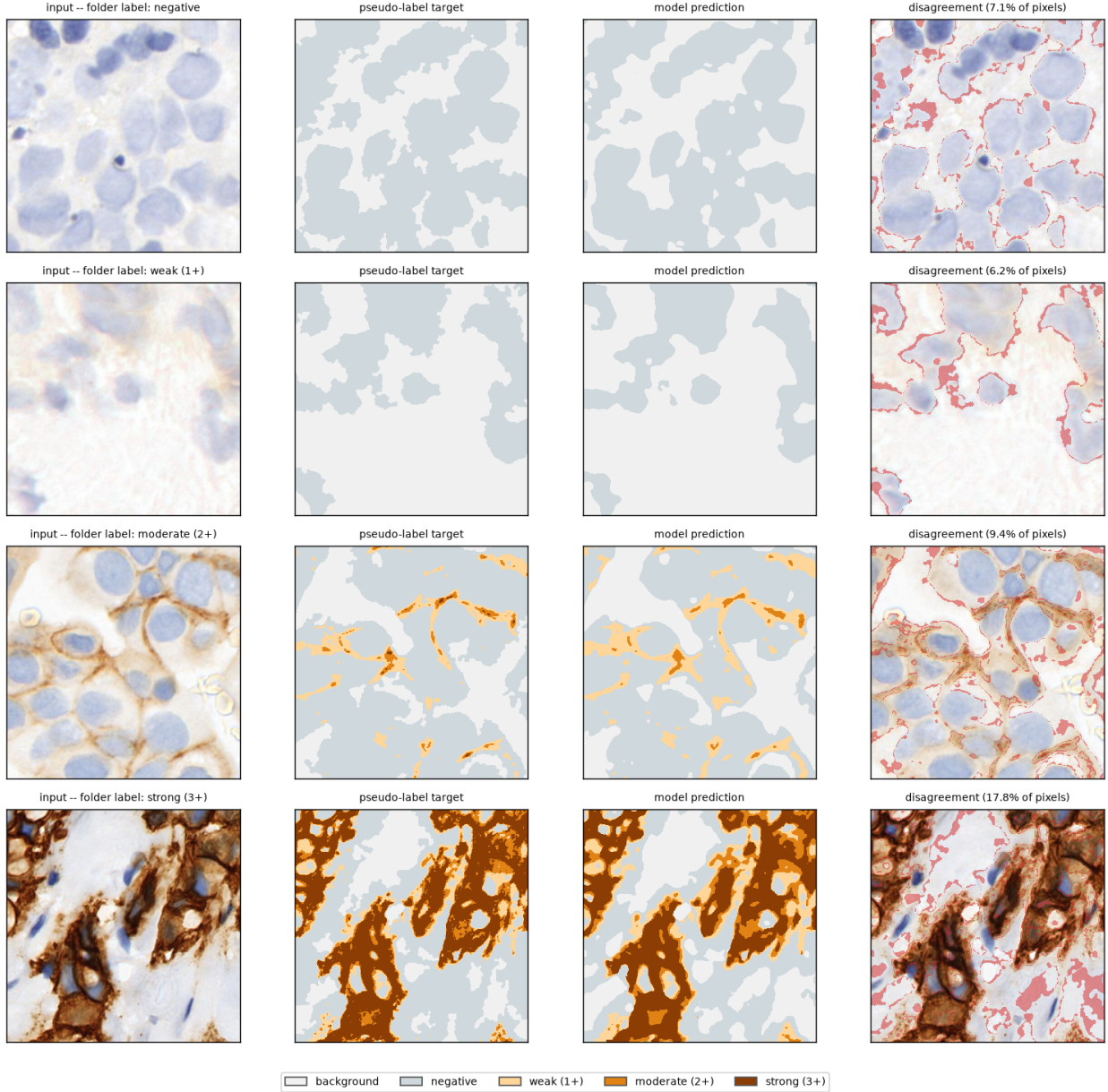


Fig. 10. Input, pseudo-label target, model prediction and pixel disagreement for one validation patch per intensity class. The model recovers thin membrane structure the pseudo-label targets are made of, rather than the smooth-blob failure diagnosed in Section IV-E.

seeds, owing to the CPU-only compute budget; single-seed results on a weakly-supervised target should be read as directional, not as precise point estimates with implied confidence intervals.

The remaining open question. Moderate (2+) is resolved as learnable (Table III), not as solved: at IoU 0.589 against 0.67–0.89 for the other stained classes, it remains the weakest class by a real margin, and it is precisely the category that decides reflex FISH testing. We conjecture, though do not yet test, that the highest-value next step

is not a further architecture search but reframing 2+ as a per-cell membrane-completeness question rather than a tissue-area intensity question – the approach ASCO/CAP itself scores by, which our pixel-area measurement only approximates.

VII. Broader Impact and Ethical Considerations

We designed this system around one non-negotiable constraint: it measures, and a pathologist scores. Every quantity this paper reports is an area percentage or a calibrated prediction set, never a categorical HER2

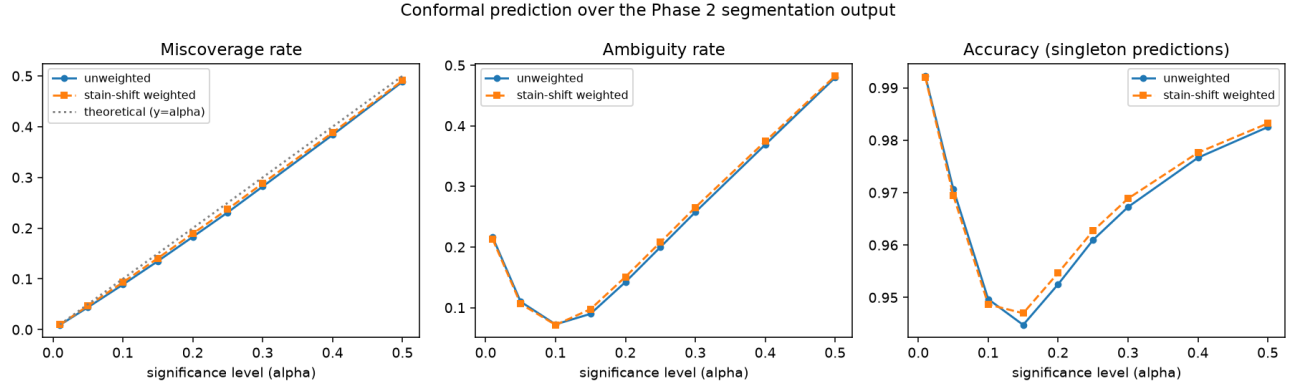


Fig. 11. Miscoverage, ambiguity and singleton accuracy across the significance-level grid, unweighted vs. stain-shift weighted, evaluated against the winning U-Net checkpoint.

call, and the surrounding system this method feeds into enforces that boundary as executable code rather than as a documentation promise – an engineering decision motivated directly by the automation-bias risk that any sufficiently accurate-looking assistive tool for a high-stakes clinical decision invites over-reliance. Conformal prediction’s role in that design is specifically to make the model’s uncertainty visible rather than to hide it behind a single confident number; a wide prediction set at a clinically relevant α is a legitimate, intended output, not a failure mode to be tuned away.

Two further considerations bear directly on responsible deployment. First, representativeness: HER2_IHC_40X’s demographic and institutional provenance is not documented at the level clinical deployment would require, and every result in this paper should be read as evidence about this dataset, not as evidence of population-level clinical validity – this is precisely the gap the stain-shift-weighted conformal machinery of Section IV-F is built to eventually address, once real multi-institution data exists to calibrate it against. Second, equity of access: by design, every component in this pipeline runs on CPU-only commodity hardware; a resource-constrained pathology department without GPU infrastructure is not, on that basis alone, excluded from using tooling of this kind. We view this as a deliberate, not incidental, property of the engineering choices in Section IV-E and Section IV-D.

VIII. Conclusion and Future Work

We show that a physically-grounded input channel (DAB optical density, computed by the same deconvolution the weak-supervision targets derive from) combined with a resolution-preserving decoder resolves a specific, diagnosed failure of a transformer segmentation baseline on HER2 IHC’s most clinically consequential class, under a strict CPU-only compute budget. We further show that an existing binary, image-level conformal prediction method for HER2 status generalizes cleanly to pixel-level, multi-class segmentation output, with stain-shift

weighting available for cross-institution deployment once genuinely multi-source data exists to exercise it. We formalized the weak-supervision setting this all sits inside (Section IV-A) precisely because, on a problem with no verified slide-level split and no pixel-level ground truth, a number reported without the estimator properties that produced it is not reproducible, however precise it looks.

Future work follows directly from Section VI: a full-scale conformal evaluation over the entire reserved holdout set; a completed inverse-frequency class-weighting comparison; multi-seed replication of the architecture comparison to bound the statistical conclusion validity threat identified above; real multi-institution data to genuinely exercise stain-shift weighting; real pathologist pixel annotations to replace pseudo-label calibration with clinical ground truth in both (1) and Section IV-F; and a per-cell membrane-completeness reformulation of the moderate (2+) class, closer to how ASCO/CAP itself defines the score.

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